

1. Equal volumes (10. mL) of 0.01M solutions of CH_3COOH and HClO were mixed and then diluted to a total volume of 100. mL. Ionization constant of CH_3COOH is equal to 1.8×10^{-5} and that for HClO is 3.7×10^{-8} .

1.1 degree of ionization for each of the acids in the solution,

1.2 degree of ionization of HClO if the diluted solution would not contain CH_3COOH ,

1.3 pH value for the solution containing at the same time CH_3COOH and HClO .

2. Coordination chemistry of an essential, multi-colouring element:

A potassium cyano complex K_1 of the element A exhibits a measured effective magnetic moment of $\mu = 3.8$ B.M.. The compound shows a d-d-absorption at 26700 cm^{-1} . The reduction of the ion $\text{A}_2\text{O}_7^{2-}$ (orange in aqueous solution) with ethanol in an acidic surrounding yields the central ion of the cyano complex. This complex K_1 may be reduced with potassium in liquid ammonia to give the cyano complex K_2 , whereby the coordination number remains unchanged. By exchanging all ligands in K_2 by ammonia, the magnetic moment of the new complex K_3 will increase to $\mu = 4.9$ B.M..

2.1 What is the element A?

2.2 Give the formula and the name of complex K_1 .

2.3 Draw the occupation of the d-orbitals for K_1 and verify it by comparing the calculated and the measured magnetic moment.

2.4 Calculate the ligand energy splitting Δ (in kJ/mol) for K_1 .

2.5 In case of the same central ion and the ligands H_2O , Cl^- and F^- the following decreasing ligand splitting energies Δ are found: 158 kJ/mol, 182 kJ/mol and 208 kJ/mol. Allocate the values to the respective ligands on the answer sheet.

2.6 Give the formula and the name of complex K_2 .

2.7 Draw the occupation of the d-orbitals for K_2 as well as K_3 and show in each case, whether it is a high-spin or a low-spin complex.

2.8 The complex with the lower oxidation number of the central ion shows a distorted octahedral structure. What is the name of this effect?

2.9 What is the geometry of the following complex of element A: $\text{A}(\text{acac})_2$? Take the fact of 2.8. into account.

3. Interconversion among Nitrogen Oxides

Nitrogen oxides (including N_2O , NO , NO_2 , N_2O_4 and etc., usually written as NO_x) are one of the main air pollutants that can cause a series of problems such as ozone depletion, acid rain, photochemical smog, and greenhouse effect. Therefore, the emission and conversion of NO_x must be controlled to improve air quality.

Here let us investigate the oxidation of NO to NO₂ via the reaction of $2\text{NO} + \text{O}_2 \rightarrow 2\text{NO}_2$.

It is generally accepted that this reaction proceeds through the following mechanism in the atmosphere:



Reaction (1) and (2) and the reverse of reaction (1) are elementary reactions. Reaction (1) is a Pre-equilibrium reaction, and its equilibrium constant in terms of concentrations is denoted as K_{c1} .

Reaction (2) is the rate-determining step of the overall reaction, and its rate constant is k_2 .

3-1: Deduce the rate expression for the overall reaction of $2\text{NO} + \text{O}_2 \rightarrow 2\text{NO}_2$ as a function of $[\text{NO}]$, $[\text{O}_2]$, K_{c1} and k_2 .

The temperature dependence of K_{c1} could be approximately described as $\ln K_{c1} = M - (N/T)$ (M and N are constants). The change of k_2 with temperature follows the Arrhenius equation with the re-exponential factor of A_2 and apparent activation energy of $E_{a,2}$. Assume that $E_{a,2}$ and A_2 are independent of the temperature.

3-2: Deduce the expressions for the pre-exponential factor (A_+) and apparent activation energy (E_{a+}) of the reaction of $2\text{NO} + \text{O}_2 \rightarrow 2\text{NO}_2$ as functions of M , N , A_2 and $E_{a,2}$.

The apparent rate constant (k_+) of the overall reaction is $6.63 \times 10^5 \text{ L}^2 \text{ mol}^{-2} \text{ min}^{-1}$ at 600 K, and its apparent activation energy is 1.20 kJ mol^{-1} .

3-3: Calculate the rate constant (in $\text{L}^2 \text{ mol}^{-2} \text{ min}^{-1}$) of this reaction at 700 K.

The standard enthalpies of formation ($\Delta_f H_m^\circ$) and standard entropies (S_m°) at 298.15 K are as follows:

	NO(g)	O ₂ (g)	NO ₂ (g)
$\Delta_f H_m^\circ (\text{kJ mol}^{-1})$	91.3		33.1
$S_m^\circ (\text{J K}^{-1} \text{ mol}^{-1})$	210.8	205.2	240.1

The standard reaction enthalpy and entropy changes of $2\text{NO} + \text{O}_2 \rightarrow 2\text{NO}_2$ can be regarded as temperature independent. All the gases are taken to be ideal for the following questions.

3-4: Calculate the thermodynamic equilibrium constant (K_p) of this reaction at 600 K.

3-5: Calculate the standard internal energy change $\Delta_r U_m^\circ$ (in kJ mol^{-1}) of this reaction at 600 K.

Experimental observation shows that the rate of the reaction of $2\text{NO}_2 \rightarrow 2\text{NO} + \text{O}_2$ is independent on the concentration of NO and O_2 .

3-6: Deduce the expression of the rate for this reaction (the apparent rate constant could be directly used as k) and calculate the value of k apparent at 600 K.

NO_2 gas is introduced to a container with a fixed volume which is held at 600 K and allowed to reach equilibrium. 20 percent of the reactant is converted to NO and O_2 . All the gases are taken to be ideal.

3-7: Calculate the total pressure of this reaction system at equilibrium.